

Praneet Raj V

📍 Bengaluru, Karnataka | 📞 +91 9108724648 | ✉ praneetraj.v@gmail.com
🐙 github.com/praneetrajv | 🔗 linkedin.com/in/praneet-raj-veera | 🌐 praneetrajv.github.io

EDUCATION

PES University B.Tech in Computer Science and Engineering | CGPA: 7.38 Expected May 2027

EXPERIENCE

Prowler Cloud Security Aug. 2026 – Present
Open Source Security Contributor Remote

- Contributed to **Prowler**, an open-source cloud security and compliance platform.
- Authored **PR #11816** implementing a security check to detect hardcoded secrets in **AWS Batch** job definitions.

Center for Information Security, Forensics and Cyber Resilience Jun. 2025 – Present
Summer Research Intern Bengaluru, India

- Developed a graph-based system for identifying and tracing the source of viral images across social media networks.
- Built an image-source detection pipeline using **SIFT**, **Perceptual Hashing (pHash)**, **NetworkX**, and temporal/network features.
- Implemented image processing and graph-analysis modules using **Python** and **OpenCV**.

PROJECTS

End-to-End Federated Learning Platform Nov. 2025 – Present

- Developing a privacy-preserving distributed learning platform that enables multiple clients to train a shared model without directly sharing their raw data.
- Implemented a **FastAPI** central server for client registration, training-round management, model distribution, and aggregation, with **REST APIs** for client communication.
- Built client-side training workflows with **Docker**, **PostgreSQL**, hardware profiling, and **JWT authentication** for secure distributed execution.

DemoGen – AI-Powered Workflow Automation Mar. 2026

- Built an AI-driven workflow automation tool using **Gemini/OpenRouter** for planning and **Selenium/Playwright** for browser interaction.
- Developed an automated video-generation pipeline using **FFmpeg/MoviePy**, with error detection and replanning for resilient execution.
- Implemented a **FastAPI/Python** backend and containerized the application using **Docker**.

Context-Aware AI Reading Copilot Aug. 2026

- Developed a **Manifest V3** Firefox ext. providing context-aware word explanations using **Google Gemini REST APIs**.
- Built asynchronous DOM extraction, dictionary lookups, client-side caching, and contextual prompt generation for real-time assistance.

ACHIEVEMENTS

Finalist – Smart India Hackathon (SIH), 2025
Top 4 – FASTATHON, SRM University All-India Hackathon
Top 10 – Health Spark Ideathon, PES University

SKILLS

Languages: Python, Java, C++, JavaScript, SQL, C **AI/ML & Data:** Federated Learning, LLM Applications, Computer Vision, SIFT, Perceptual Hashing, Network Analysis **Frameworks:** FastAPI, React.js, OpenCV, NetworkX, Playwright, Selenium **Databases & Tools:** PostgreSQL, MongoDB, Docker, Git, GitHub, Linux, REST APIs

CERTIFICATIONS

Oracle Cloud Infrastructure AI Foundations Associate | Data Platform Foundations Associate
Redis Associate Software Operator